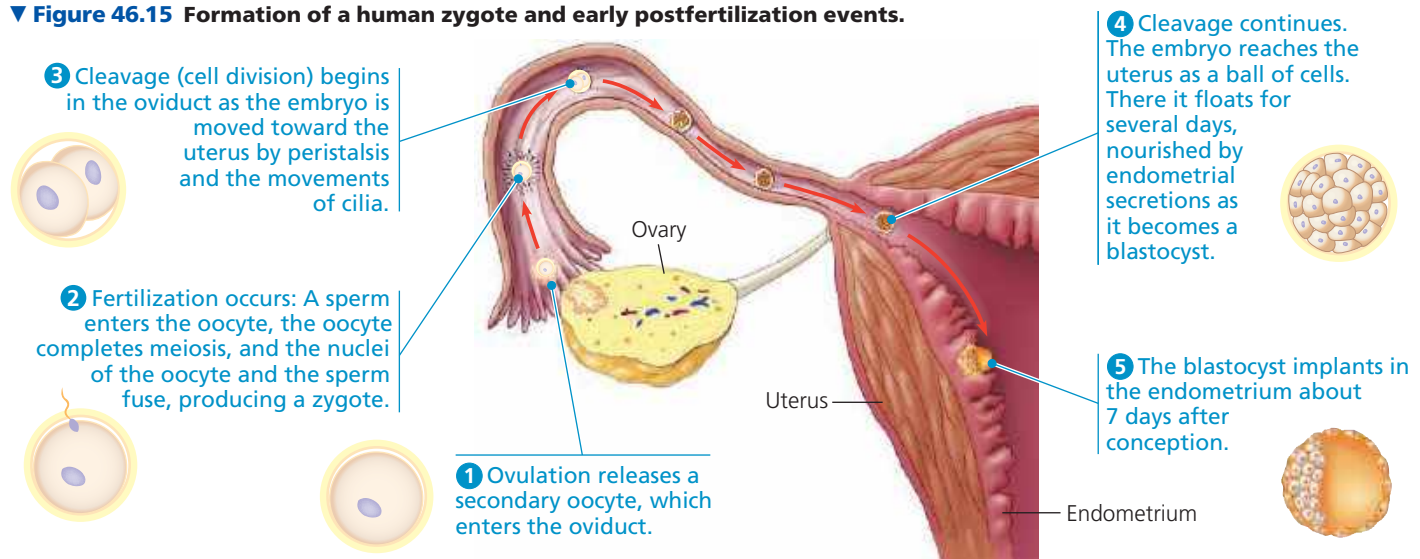


▼ **Figure 46.15** Formation of a human zygote and early postfertilization events.



**VISUAL SKILLS** If a woman's eggs need to be fertilized *in vitro*, they can be readily introduced into the uterus but not the extremely narrow oviduct. Based on this drawing, propose conditions for culturing a fertilized egg that you predict will optimize the chance of a successful pregnancy.

rodents, 280 days in cows, and more than 600 days in elephants. The roughly nine months of human gestation are divided into three *trimesters* of equal length.

### First Trimester

During the first trimester, the implanted embryo secretes hormones that signal its presence and regulate the mother's reproductive system. One embryonic hormone, *human chorionic gonadotropin (hCG)*, acts like pituitary LH in

maintaining secretion of progesterone and estrogens by the corpus luteum through the first few months of pregnancy. Some hCG passes from the maternal blood to the urine, where it can be detected by the most common early pregnancy tests.

During its first 2–4 weeks of development, the embryo obtains nutrients directly from the endometrium. Meanwhile, the outer layer of the blastocyst, which is called the **trophoblast**, grows outward and mingles with the endometrium, eventually helping form the **placenta** (**Figure 46.16**).

► **Figure 46.16** Placental

**circulation.** Maternal blood enters the placenta in arteries, flows through blood pools in the endometrium, and leaves via veins. Embryonic or fetal blood, which remains in vessels, enters the placenta through arteries and passes through capillaries in finger-like chorionic villi, where oxygen and nutrients are acquired. Fetal blood leaves the placenta through veins leading back to the fetus.

**WHAT IF?** In a rare genetic disorder, the absence of a particular enzyme leads to increased testosterone production. When the fetus has this disorder, the mother develops a male-like pattern of body hair during the pregnancy. Explain why.

